



Methods

Audio segmentation

- AE-MRI, AE-EMA & FR-MRI came w/ manual word-level & phoneme-level segmentation.
- BE-US was segmented through Montreal Forced Aligner (McAuliffe et al. 2017).
- The phonetic transcriptions were then mapped back to the underlying phonemic transcriptions based on *espeak* & *eng_to_ipa*.





Methods

Preview

1. Data: acoustic/articulatory data from published datasets
- 2. Distinctiveness measurement in acoustics & articulation**
3. Bigram surprisal calculation
4. Phoneme flexibility: degree of freedom between articulatory synergies & the speech target



USC